

Noether's Normalization and Homotopy Continuation: Solving Positive Dimensional Systems

Vincenzo Antonio Isoldi
M.Sc. Student in Mathematics

Department of Mathematics
University of Trento

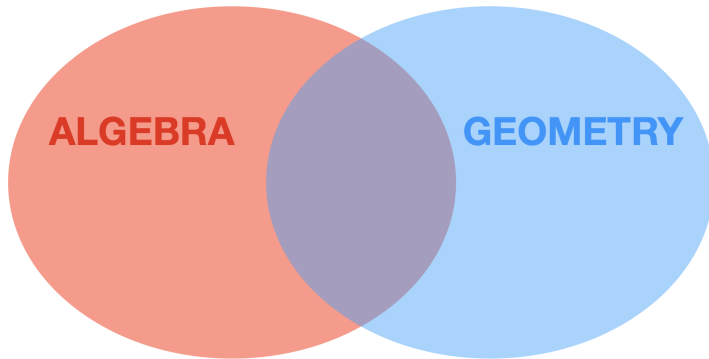


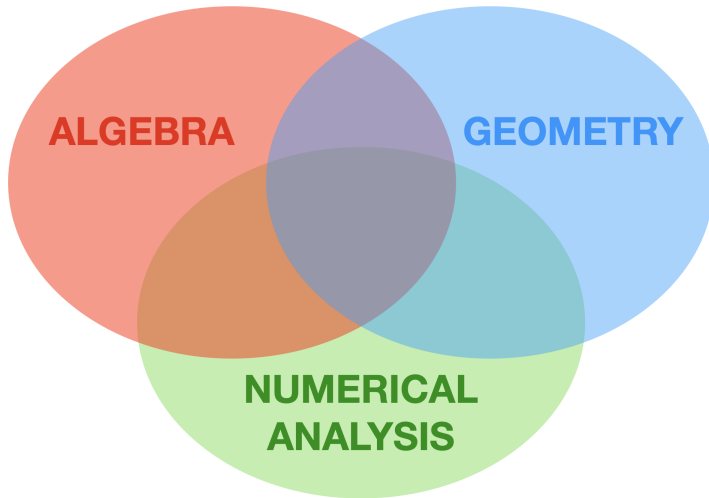
What is our problem?

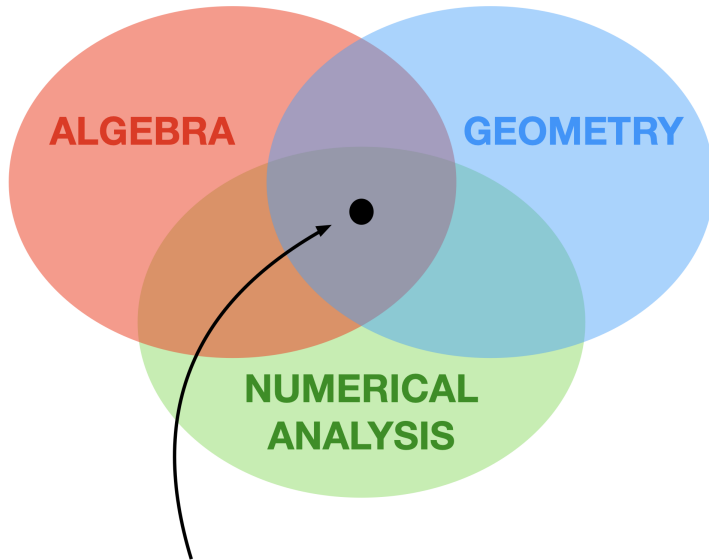
What is our problem?

Given a system of polynomials $f_1(\mathbf{x}), \dots, f_n(\mathbf{x}) = 0$, find all possible solutions.

ALGEBRA







HOMOTOPY CONTINUATION

Homotopy Continuation: An Intuition

Say we want to solve the following polynomial system:

$$F : \begin{cases} x^2 + 4y^2 - 4 = 0, \\ 2y^2 - x = 0. \end{cases}$$

The symbolic approach gives the lexicographic Gröbner basis:

$$(2y^2 - x, x^2 + 2x - 4)$$

Hence, the solutions are easily computable.

However, computing Gröbner basis is in general a really hard problem.

Homotopy Continuation: An Intuition

Say we want to solve the following polynomial system:

$$F : \begin{cases} x^2 + 4y^2 - 4 = 0, \\ 2y^2 - x = 0. \end{cases}$$

The symbolic approach gives the lexicographic Gröbner basis:

$$(2y^2 - x, x^2 + 2x - 4)$$

Hence, the solutions are easily computable.

However, computing Gröbner basis is in general a really hard problem.

Can we approach the same problem differently, perhaps exploiting numerical methods?

Homotopy Continuation: An Intuition

We want to solve:

$$F : \begin{cases} x^2 + 4y^2 - 4 = 0, \\ 2y^2 - x = 0. \end{cases}$$

Homotopy Continuation: An Intuition

We want to solve:

$$F : \begin{cases} x^2 + 4y^2 - 4 = 0, \\ 2y^2 - x = 0. \end{cases}$$

Idea: We consider the easier polynomial system:

$$G : \begin{cases} x^2 - 1 = 0, \\ y^2 - 1 = 0. \end{cases}$$

Homotopy Continuation: An Intuition

We want to solve:

$$F : \begin{cases} x^2 + 4y^2 - 4 = 0, \\ 2y^2 - x = 0. \end{cases}$$

Idea: We consider the easier polynomial system:

$$G : \begin{cases} x^2 - 1 = 0, \\ y^2 - 1 = 0. \end{cases}$$

and we fit them in a homotopy:

$$H_t = H(x, y; t) = (1 - t)F + tG$$

Homotopy Continuation: An Intuition

We want to solve:

$$F : \begin{cases} x^2 + 4y^2 - 4 = 0, \\ 2y^2 - x = 0. \end{cases}$$

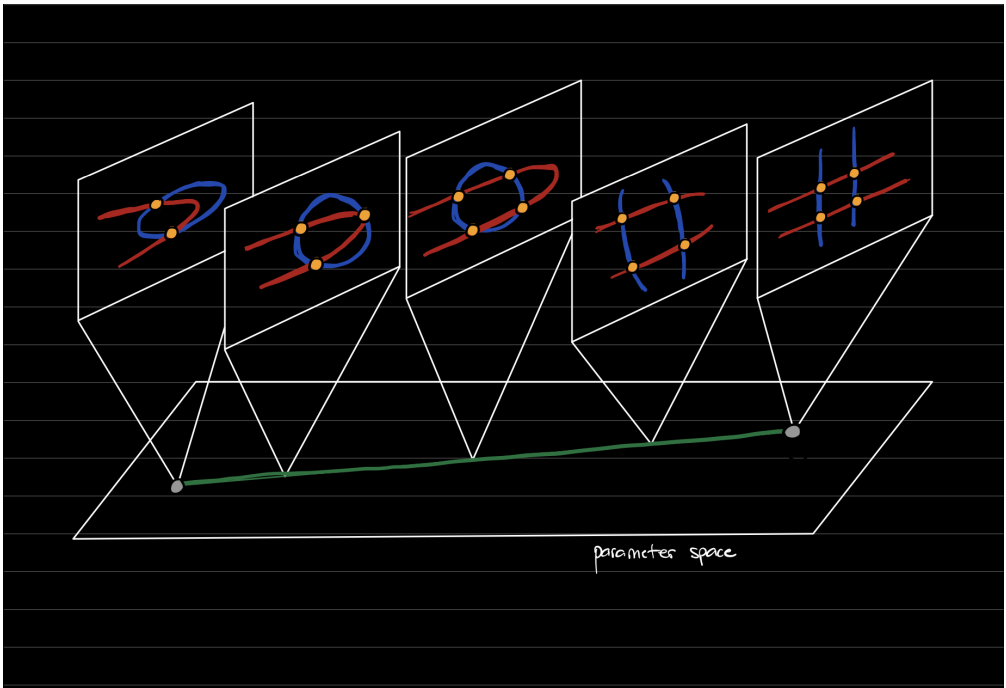
Idea: We consider the easier polynomial system:

$$G : \begin{cases} x^2 - 1 = 0, \\ y^2 - 1 = 0. \end{cases}$$

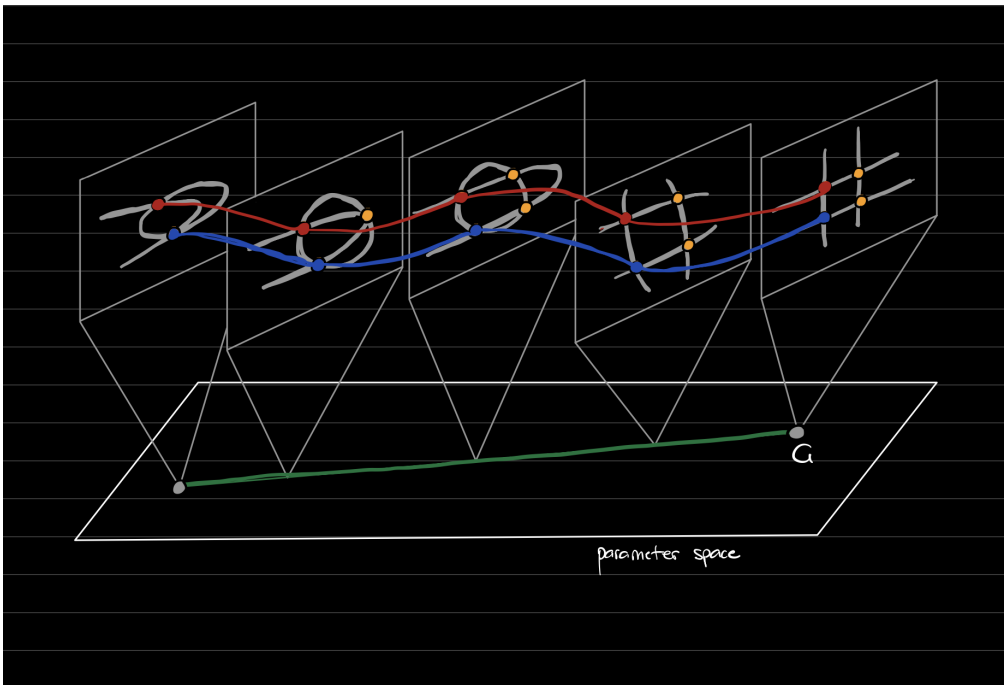
and we fit them in a homotopy:

$$H_t = H(x, y; t) = (1 - t)F + tG$$

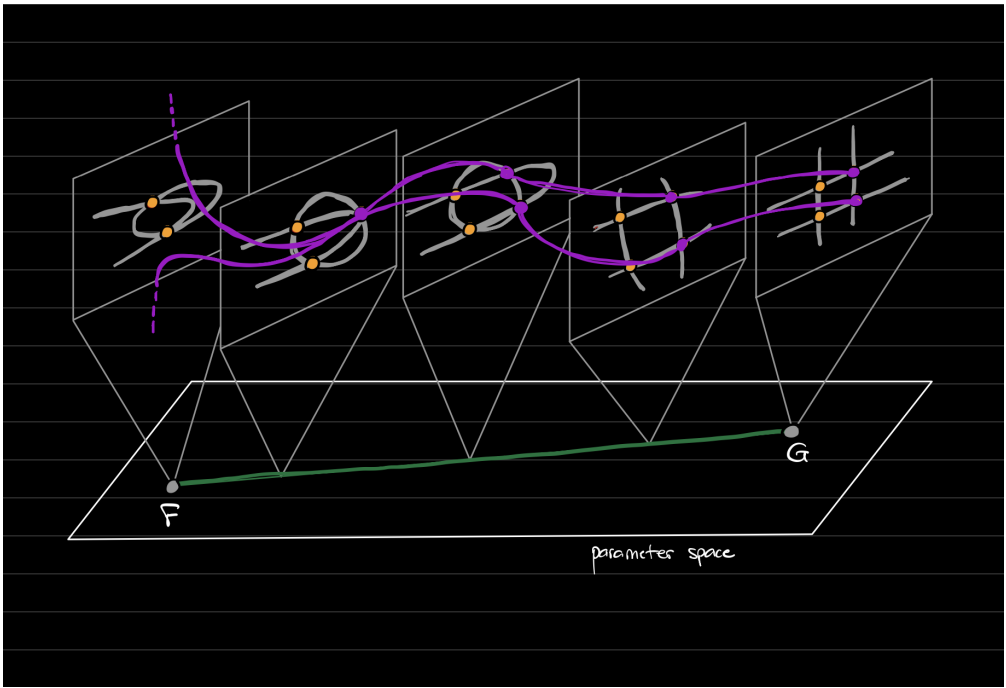
Then, we **"track"** the solutions among the homotopy, starting from the known solutions of $G = 0$.



Courtesy of Prof. A. Oneto - GTDA 2024



Courtesy of Prof. A. Oneto - GTDA 2024



Courtesy of Prof. A. Oneto - GTDA 2024

How to track the paths?

Assume $\mathbf{z}^*(t)$ is a solution path. Then:

$$\begin{aligned} 0 = \frac{dH(\mathbf{z}^*(t), t)}{dt} &= \frac{\partial H}{\partial \mathbf{z}}(\mathbf{z}^*(t), t) \frac{d\mathbf{z}^*(t)}{dt} + \frac{\partial H}{\partial t}(\mathbf{z}^*(t), t) \\ \implies \frac{d\mathbf{z}^*(t)}{dt} &= -\frac{\frac{\partial H}{\partial t}(\mathbf{z}^*(t), t)}{\frac{\partial H}{\partial \mathbf{z}}(\mathbf{z}^*(t), t)} \end{aligned}$$

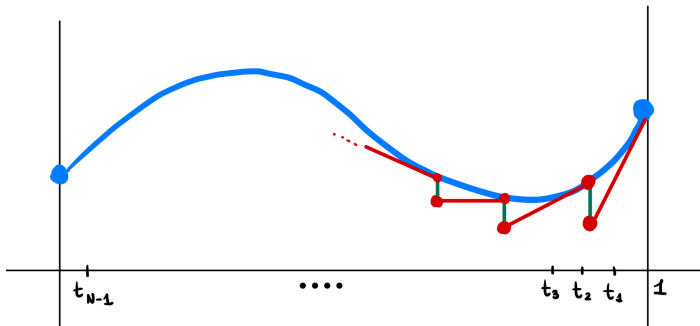
This gives an ordinary differential equation for $\mathbf{z}^*(t)$ which can be solved numerically by considering the starting point $\mathbf{z}^*(1)$.

How to track the paths?

Assume $\mathbf{z}^*(t)$ is a solution path. Then:

$$0 = \frac{dH(\mathbf{z}^*(t), t)}{dt} = \frac{\partial H}{\partial \mathbf{z}}(\mathbf{z}^*(t), t) \frac{d\mathbf{z}^*(t)}{dt} + \frac{\partial H}{\partial t}(\mathbf{z}^*(t), t)$$
$$\Rightarrow \frac{d\mathbf{z}^*(t)}{dt} = -\frac{\frac{\partial H}{\partial t}(\mathbf{z}^*(t), t)}{\frac{\partial H}{\partial \mathbf{z}}(\mathbf{z}^*(t), t)}$$

This gives an ordinary differential equation for $\mathbf{z}^*(t)$ which can be solved numerically by considering the starting point $\mathbf{z}^*(1)$.



What if we don't have finitely many solutions?

What if our solution is one (or more!) variety of positive dimension?

PART 1: THE THEORY

NOETHER'S NORMALIZATION LEMMA

Part 1: The Theory

1 Finite Ring Extension

- Finite and Integral A -algebras
- Finite vs Integral
- Integral Closure

2 Noether Normalization

- Preliminary Results
- Noether Normalization Lemma
- Consequences of the Theorem
- Algebraic and Geometric Interpretation

Rings behave worse than Fields

We know that, if $K \subset L$ is a field extension, an element $y \in L$ is *algebraic* over K if it satisfies an algebraic dependence relation

$$f(y) = a_n y^n + \cdots + a_1 y + a_0 = 0,$$

with some non-zero polynomial $f \in K[Y]$; without loss of generality, $a_n \neq 0$. Over a field, it costs nothing to divide the relation through by a_n , so we always just assume that $a_n = 1$, say that f is *monic*, and think no more about it.

Rings behave worse than Fields

We know that, if $K \subset L$ is a field extension, an element $y \in L$ is *algebraic* over K if it satisfies an algebraic dependence relation

$$f(y) = a_n y^n + \cdots + a_1 y + a_0 = 0,$$

with some non-zero polynomial $f \in K[Y]$; without loss of generality, $a_n \neq 0$. Over a field, it costs nothing to divide the relation through by a_n , so we always just assume that $a_n = 1$, say that f is *monic*, and think no more about it.

Over a more general ring A , however, you can't just divide by a_n . A relation

$$f(y) = y^n + a_{n-1}y^{n-1} + \cdots + a_1 y + a_0 = 0$$

with a monic polynomial $f \in A[Y]$ is called an *integral* dependence relation for y ; the distinction between algebraic dependence and integral dependence is crucial. This notion leads at once to the ring of integers O_K of a number field K , which is the basic definition of algebraic number theory.

Rings behave worse than Fields

We know that, if $K \subset L$ is a field extension, an element $y \in L$ is *algebraic* over K if it satisfies an algebraic dependence relation

$$f(y) = a_n y^n + \cdots + a_1 y + a_0 = 0,$$

with some non-zero polynomial $f \in K[Y]$; without loss of generality, $a_n \neq 0$. Over a field, it costs nothing to divide the relation through by a_n , so we always just assume that $a_n = 1$, say that f is *monic*, and think no more about it.

Over a more general ring A , however, you can't just divide by a_n . A relation

$$f(y) = y^n + a_{n-1}y^{n-1} + \cdots + a_1 y + a_0 = 0$$

with a monic polynomial $f \in A[Y]$ is called an *integral* dependence relation for y ; the distinction between algebraic dependence and integral dependence is crucial. This notion leads at once to the ring of integers O_K of a number field K , which is the basic definition of algebraic number theory.

Integral dependence

Rings behave worse than Fields

We know that, if $K \subset L$ is a field extension, an element $y \in L$ is *algebraic* over K if it satisfies an algebraic dependence relation

$$f(y) = a_n y^n + \cdots + a_1 y + a_0 = 0,$$

with some non-zero polynomial $f \in K[Y]$; without loss of generality, $a_n \neq 0$. Over a field, it costs nothing to divide the relation through by a_n , so we always just assume that $a_n = 1$, say that f is *monic*, and think no more about it.

Over a more general ring A , however, you can't just divide by a_n . A relation

$$f(y) = y^n + a_{n-1}y^{n-1} + \cdots + a_1 y + a_0 = 0$$

with a monic polynomial $f \in A[Y]$ is called an *integral* dependence relation for y ; the distinction between algebraic dependence and integral dependence is crucial. This notion leads at once to the ring of integers O_K of a number field K , which is the basic definition of algebraic number theory.

Integral dependence $\leftrightarrow$ Finiteness conditions

Rings behave worse than Fields

We know that, if $K \subset L$ is a field extension, an element $y \in L$ is *algebraic* over K if it satisfies an algebraic dependence relation

$$f(y) = a_n y^n + \cdots + a_1 y + a_0 = 0,$$

with some non-zero polynomial $f \in K[Y]$; without loss of generality, $a_n \neq 0$. Over a field, it costs nothing to divide the relation through by a_n , so we always just assume that $a_n = 1$, say that f is *monic*, and think no more about it.

Over a more general ring A , however, you can't just divide by a_n . A relation

$$f(y) = y^n + a_{n-1}y^{n-1} + \cdots + a_1 y + a_0 = 0$$

with a monic polynomial $f \in A[Y]$ is called an *integral* dependence relation for y ; the distinction between algebraic dependence and integral dependence is crucial. This notion leads at once to the ring of integers O_K of a number field K , which is the basic definition of algebraic number theory.

Integral dependence $\leftrightarrow$ Finiteness conditions $\leftrightarrow$ Noether Normalization

1 Finite Ring Extension

- Finite and Integral A -algebras
- Finite vs Integral
- Integral Closure

2 Noether Normalization

- Preliminary Results
- Noether Normalization Lemma
- Consequences of the Theorem
- Algebraic and Geometric Interpretation

Finite and Integral A -algebras

Let A be a ring; an A -algebra is by definition a ring with a given ring homomorphism $\phi : A \rightarrow B$. The point is that B is then an A -module, with the multiplication defined by $\phi(a) \cdot b$. An important case of this is when $B = A$, when B is also called an extension ring of A ; we can usually reduce to this case by writing $\phi(A) = A' \subset B$.

Let A be a ring; an A -algebra is by definition a ring with a given ring homomorphism $\phi : A \rightarrow B$. The point is that B is then an A -module, with the multiplication defined by $\phi(a) \cdot b$. An important case of this is when $B = A$, when B is also called an extension ring of A ; we can usually reduce to this case by writing $\phi(A) = A' \subset B$.

Definition

Let B be an A -algebra.

- (i) B is a finite A -algebra (or is finite over A) if it is finite as an A -module.
- (ii) An element $b \in B$ is *integral over A* if there exists a monic polynomial $f(Y) = Y^n + a_{n-1}Y^{n-1} + \cdots + a_0 \in A[Y]$ such that

$$f(b) = b^n + a_{n-1}b^{n-1} + \cdots + a_0 = 0. \quad (1)$$

The algebra B is integral over A if every $b \in B$ is integral.

- (i) $\mathbb{Z} \left[\frac{1}{3} \right]$ is not integral over $\mathbb{Z}$, and for $f \in k[X]$ an irreducible polynomial, $k[X] \left[\frac{1}{f} \right]$ is not integral over $k[X]$; more generally, if A is a UFD with field of fractions K , then no element of $K \setminus A$ is integral over A .

- (i) $\mathbb{Z} \left[\frac{1}{3} \right]$ is not integral over $\mathbb{Z}$, and for $f \in k[X]$ an irreducible polynomial, $k[X] \left[\frac{1}{f} \right]$ is not integral over $k[X]$; more generally, if A is a UFD with field of fractions K , then no element of $K \setminus A$ is integral over A .
- (ii) $k[X^2] \subseteq k[X]$ is an integral extension.

- (i) $\mathbb{Z}[\frac{1}{3}]$ is not integral over $\mathbb{Z}$, and for $f \in k[X]$ an irreducible polynomial, $k[X][\frac{1}{f}]$ is not integral over $k[X]$; more generally, if A is a UFD with field of fractions K , then no element of $K \setminus A$ is integral over A .
- (ii) $k[X^2] \subseteq k[X]$ is an integral extension.
- (iii) $\mathbb{Z} \subseteq \mathbb{Z}[\varphi]$ is integral, where $\varphi = \frac{1+\sqrt{5}}{2}$ is the "golden ratio", because $\varphi^2 - \varphi - 1 = 0$; on the other hand, $\alpha = \frac{1+\sqrt{3}}{2}$ is not integral over $\mathbb{Z}$ (because $\alpha^2 - \alpha = \frac{1}{2}$).

1 Finite Ring Extension

- Finite and Integral A -algebras
- Finite vs Integral
- Integral Closure

2 Noether Normalization

- Preliminary Results
- Noether Normalization Lemma
- Consequences of the Theorem
- Algebraic and Geometric Interpretation

Finite vs Integral and Tower Laws

Proposition (Finite vs Integral)

Let $\varphi : A \rightarrow B$ be an A -algebra, and $y \in B$; then three equivalent conditions:

- (i) y is integral over A .
- (ii) The subring $A'[y] \subset B$ generated by $A' = \varphi(A)$ and y is finite over A .
- (iii) There exists an A -subalgebra $C \subset B$ such that $A'[y] \subset C$ and C is finite over A .

From now on we only deal with extension rings $A \subset B$.

Finite vs Integral and Tower Laws

Proposition (Finite vs Integral)

Let $\varphi : A \rightarrow B$ be an A -algebra, and $y \in B$; then three equivalent conditions:

- (i) y is integral over A .
- (ii) The subring $A'[y] \subset B$ generated by $A' = \varphi(A)$ and y is finite over A .
- (iii) There exists an A -subalgebra $C \subset B$ such that $A'[y] \subset C$ and C is finite over A .

From now on we only deal with extension rings $A \subset B$.

Proposition (Tower laws)

Let B be an A -algebra.

- (a) If $A \subset B \subset C$ are extension rings such that C is a finite B -algebra and B a finite A -algebra, then C is finite over A .
- (b) If $y_1, \dots, y_m \in B$ are integral over A then $A[y_1, \dots, y_m]$ is finite over A ; in particular, every $f \in A[y_1, \dots, y_m]$ is integral over A .
- (c) If $A \subset B \subset C$ with C integral over B and B integral over A then C is integral over A .
- (d) The subset $\tilde{A} = \{y \in B \mid y \text{ is integral over } A\} \subset B$ is a subring of B ; moreover, if $y \in B$ is integral over $\tilde{A}$ then $y \in \tilde{A}$, so that $\tilde{\tilde{A}} = \tilde{A}$

1 Finite Ring Extension

- Finite and Integral A -algebras
- Finite vs Integral
- Integral Closure

2 Noether Normalization

- Preliminary Results
- Noether Normalization Lemma
- Consequences of the Theorem
- Algebraic and Geometric Interpretation

Definition

The ring $\tilde{A}$ appearing in the previous Proposition, (d), is the *integral closure* of A in B ; if $A = \tilde{A}$ then A is *integrally closed* in B . An integral domain A is *normal* if it is integrally closed in its field of fractions, that is, if $A = \tilde{A} \subset K = \text{Frac } A$; for any integral domain A , the integral closure of A in its field of fractions $K = \text{Frac } A$ is also called the *normalization* of A .

Definition

The ring $\tilde{A}$ appearing in the previous Proposition, (d), is the *integral closure* of A in B ; if $A = \tilde{A}$ then A is *integrally closed* in B . An integral domain A is *normal* if it is integrally closed in its field of fractions, that is, if $A = \tilde{A} \subset K = \text{Frac } A$; for any integral domain A , the integral closure of A in its field of fractions $K = \text{Frac } A$ is also called the *normalization* of A .

An important particular case of this definition is the ring of integers in algebraic number theory: a *number field* is a finite field extension $\mathbb{Q} \subset K$ of the rational field (that is, $[K : \mathbb{Q}] = \dim_{\mathbb{Q}} K < \infty$). The *ring of integers* O_K of K is the ring $\{y \in K \mid y \text{ is integral over } \mathbb{Z}\}$, the integral closure of $\mathbb{Z}$ in K .

(i) A UFD is normal.

- (i) A UFD is normal.
- (ii) Consider the quadratic number field $\mathbb{Q} \subset \mathbb{Q}(\sqrt{n})$, where n is a square-free integer, and the subring $\mathbb{Z} \subset \mathbb{Q}[\sqrt{n}]$. Set

$$\begin{cases} \alpha = (1 + \sqrt{n})/2 & \text{if } n \equiv 1 \pmod{4} \\ \alpha = \sqrt{n} & \text{if } n \equiv 2 \text{ or } 3 \pmod{4} \end{cases}$$

then it is not hard to see that the integral closure of $\mathbb{Z}$ in $\mathbb{Q}(\sqrt{n})$ is $\mathbb{Z}[\alpha]$.

- (i) A UFD is normal.
- (ii) Consider the quadratic number field $\mathbb{Q} \subset \mathbb{Q}(\sqrt{n})$, where n is a square-free integer, and the subring $\mathbb{Z} \subset \mathbb{Q}[\sqrt{n}]$. Set

$$\begin{cases} \alpha = (1 + \sqrt{n})/2 & \text{if } n \equiv 1 \pmod{4} \\ \alpha = \sqrt{n} & \text{if } n \equiv 2 \text{ or } 3 \pmod{4} \end{cases}$$

then it is not hard to see that the integral closure of $\mathbb{Z}$ in $\mathbb{Q}(\sqrt{n})$ is $\mathbb{Z}[\alpha]$.

- (iii) Consider the ring $A = k[X, Y]/(Y^2 - X^3)$, and write $x, y \in A$ for the classes of X, Y ; then A is not normal: it is not hard to see that it has field of fractions $\text{Frac } A = k(t)$, where $t = y/x$, and $x = t^2, y = t^3$; either of these relations show that t is integral over A , but obviously $t \notin A$. Also, $k[t]$ is normal (because it's a UFD), so it is the integral closure of A in $k(t)$.

1 Finite Ring Extension

- Finite and Integral A -algebras
- Finite vs Integral
- Integral Closure

2 Noether Normalization

- Preliminary Results
- Noether Normalization Lemma
- Consequences of the Theorem
- Algebraic and Geometric Interpretation

Preliminary Results

For the rest of this section, k is a field, and A is a k -algebra.

Preliminary Results

For the rest of this section, k is a field, and A is a k -algebra.

Definition

Elements $y_1, \dots, y_n \in A$ are algebraically independent over k if the natural surjection $k[Y_1, \dots, Y_n] \rightarrow k[y_1, \dots, y_n]$ is an isomorphism, where the left-hand side is the polynomial ring. This just means that there are no nonzero polynomial relations $F(y_1, \dots, y_n) = 0$ with coefficients in k .

Preliminary Results

For the rest of this section, k is a field, and A is a k -algebra.

Definition

Elements $y_1, \dots, y_n \in A$ are algebraically independent over k if the natural surjection $k[Y_1, \dots, Y_n] \rightarrow k[y_1, \dots, y_n]$ is an isomorphism, where the left-hand side is the polynomial ring. This just means that there are no nonzero polynomial relations $F(y_1, \dots, y_n) = 0$ with coefficients in k .

Recall that a k -algebra A is finitely generated (f.g.) over k if

$$A = \frac{k[y_1, \dots, y_n]}{I}$$

for some $I \subset k[y_1, \dots, y_n]$ ideal.

Main Claim

Suppose that $A = k[y_1, \dots, y_n]$, and that

$$0 \neq F \in k[Y_1, \dots, Y_n]$$

is such that $F(y_1, \dots, y_n) = 0$. Then there exists $y_1^*, \dots, y_{n-1}^* \in A$ such that y_n is integral over $A^* = k[y_1^*, \dots, y_{n-1}^*]$ and $A = A^*[y_n]$.

Main Claim

Suppose that $A = k[y_1, \dots, y_n]$, and that

$$0 \neq F \in k[Y_1, \dots, Y_n]$$

is such that $F(y_1, \dots, y_n) = 0$. Then there exists $y_1^*, \dots, y_{n-1}^* \in A$ such that y_n is integral over $A^* = k[y_1^*, \dots, y_{n-1}^*]$ and $A = A^*[y_n]$.

There exist several proofs of this claim. A general one is due to Nagata in the 1950s. There is an alternative proof traditional in algebraic geometry, which works if the field k is infinite. Let $\alpha_1, \dots, \alpha_{n-1} \in k$, and set

$$y_i^* = y_i - \alpha_i y_n \quad \text{for } i = 1, \dots, n-1$$

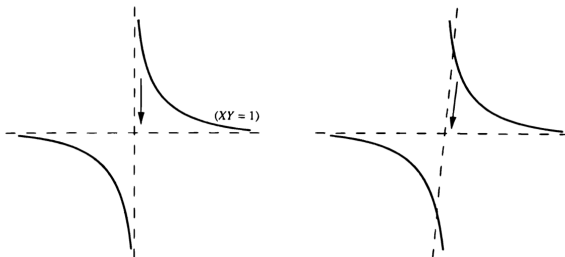
Then it's not hard to see that if k is infinite, the α_i can be chosen so that $F(y_i^* + \alpha_i y_n, y_n) = 0$ is a monic relation for y_n over $k[y_1^*, \dots, y_{n-1}^*]$. See any textbook on algebraic geometry, for example, [UAG], §3.

The Geometry of the Proof

To illustrate the above proof, consider $A = k[X, Y]/(XY - 1)$. Then Y is algebraic over $k[X]$ (in fact, $Y = X^{-1} \in k(X)$), but not integral over $k[X]$. This corresponds to the fact that the hyperbola $XY = 1$ has the line $X = 0$ as an asymptotic line, so that its projection to the X -axis has a "missing" root over $X = 0$.

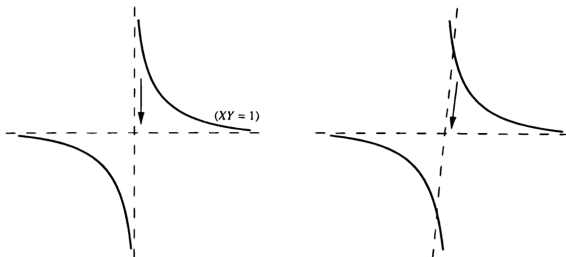
The Geometry of the Proof

To illustrate the above proof, consider $A = k[X, Y]/(XY - 1)$. Then Y is algebraic over $k[X]$ (in fact, $Y = X^{-1} \in k(X)$), but not integral over $k[X]$. This corresponds to the fact that the hyperbola $XY = 1$ has the line $X = 0$ as an asymptotic line, so that its projection to the X -axis has a "missing" root over $X = 0$. If I take $X' = X - \varepsilon Y$ as the basic transcendental element of A instead of X , the relation becomes $(X' + \varepsilon Y)Y = 1$, which is monic in Y if $\varepsilon \neq 0$. This corresponds geometrically to tilting the hyperbola a little before projecting, so that it no longer has a vertical asymptotic line.



The Geometry of the Proof

To illustrate the above proof, consider $A = k[X, Y]/(XY - 1)$. Then Y is algebraic over $k[X]$ (in fact, $Y = X^{-1} \in k(X)$), but not integral over $k[X]$. This corresponds to the fact that the hyperbola $XY = 1$ has the line $X = 0$ as an asymptotic line, so that its projection to the X -axis has a "missing" root over $X = 0$. If I take $X' = X - \varepsilon Y$ as the basic transcendental element of A instead of X , the relation becomes $(X' + \varepsilon Y)Y = 1$, which is monic in Y if $\varepsilon \neq 0$. This corresponds geometrically to tilting the hyperbola a little before projecting, so that it no longer has a vertical asymptotic line.



The moral of either proof is that for any choice $y_1, \dots, y_{n-1}$, we'll be fairly unlucky if y_n is not integral over $k[y_1, \dots, y_{n-1}]$. Perturbing the choice of $y_1, \dots, y_{n-1}$, I can guarantee to avoid the unlucky accident.

1 Finite Ring Extension

- Finite and Integral A -algebras
- Finite vs Integral
- Integral Closure

2 Noether Normalization

- Preliminary Results
- **Noether Normalization Lemma**
- Consequences of the Theorem
- Algebraic and Geometric Interpretation

Noether Normalization Lemma

Theorem

Let k be a field, and A a f.g. k -algebra; then there exist elements $z_1, \dots, z_m \in A$ such that:

- (i) $z_1, \dots, z_m$ are algebraically independent over k ;
- (ii) A is finite over $B = k[z_1, \dots, z_m]$.

That is, a f.g. extension $k \subset A$ can be written as a composite

$$k \subset B = k[z_1, \dots, z_m] \subset A$$

where $k \subset B$ is a polynomial extension, and $B \subset A$ is finite. This is a rough structure theorem for finitely generated k -algebras.

Proof (assuming the claim)

Proof. This is an easy induction on the number n of generators of A : if $n = 0$, there's nothing to prove. If $n > 0$ and $y_1, \dots, y_n$ are algebraically independent over k , then again there's nothing to prove. So suppose that $y_1, \dots, y_n$ are algebraically dependent over k , and let $0 \neq F \in k[Y_1, \dots, Y_n]$ be such that $F(y_1, \dots, y_n) = 0$; then, by the claim, there exist $y_1^*, \dots, y_{n-1}^* \in A$ such that y_n is integral over $A^* = k[y_1^*, \dots, y_{n-1}^*]$ and $A = A^*[y_n]$.

Proof (assuming the claim)

Proof. This is an easy induction on the number n of generators of A : if $n = 0$, there's nothing to prove. If $n > 0$ and $y_1, \dots, y_n$ are algebraically independent over k , then again there's nothing to prove. So suppose that $y_1, \dots, y_n$ are algebraically dependent over k , and let $0 \neq F \in k[Y_1, \dots, Y_n]$ be such that $F(y_1, \dots, y_n) = 0$; then, by the claim, there exist $y_1^*, \dots, y_{n-1}^* \in A$ such that y_n is integral over $A^* = k[y_1^*, \dots, y_{n-1}^*]$ and $A = A^*[y_n]$.

Now by the inductive hypothesis applied to A^* , there exist elements $z_1, \dots, z_m \in A^*$ that are algebraically independent over k and with A^* finite over $B = k[z_1, \dots, z_m]$. Now since y_n is integral over A^* , it follows that $A^*[y_n]$ is finite over A^* , so that each step $B \subset A^* \subset A^*[y_n] = A$ is finite, and A is finite over B , as required. *Q.E.D.*

1 Finite Ring Extension

- Finite and Integral A -algebras
- Finite vs Integral
- Integral Closure

2 Noether Normalization

- Preliminary Results
- Noether Normalization Lemma
- Consequences of the Theorem
- Algebraic and Geometric Interpretation

Zariski's Lemma

Corollary

Let R be a finitely generated algebra over a field K , and suppose that R is a field. Then R is a finite algebraic extension of K , i.e., R is module-finite over K .

Zariski's Lemma

Corollary

Let R be a finitely generated algebra over a field K , and suppose that R is a field. Then R is a finite algebraic extension of K , i.e., R is module-finite over K .

Proof. By the Noether Normalization Lemma, R is module-finite over some polynomial subring $K[z_1, \dots, z_d]$. If $d \geq 1$, the polynomial ring has dimension of at least one, and then R has dimension of at least one, a contradiction. Thus, $d = 0$, and R is module-finite over K . Since R is a field, this means precisely that R is a finite algebraic extension of K . *Q.E.D.*

Zariski's Lemma

Corollary

Let R be a finitely generated algebra over a field K , and suppose that R is a field. Then R is a finite algebraic extension of K , i.e., R is module-finite over K .

Proof. By the Noether Normalization Lemma, R is module-finite over some polynomial subring $K[z_1, \dots, z_d]$. If $d \geq 1$, the polynomial ring has dimension of at least one, and then R has dimension of at least one, a contradiction. Thus, $d = 0$, and R is module-finite over K . Since R is a field, this means precisely that R is a finite algebraic extension of K . *Q.E.D.*

Corollary

Let K be an algebraically closed field, let R be a finitely generated K -algebra, and let m be a maximal ideal of R . Then the composite map $K \rightarrow R \rightarrow R/m$ is an isomorphism.

Zariski's Lemma

Corollary

Let R be a finitely generated algebra over a field K , and suppose that R is a field. Then R is a finite algebraic extension of K , i.e., R is module-finite over K .

Proof. By the Noether Normalization Lemma, R is module-finite over some polynomial subring $K[z_1, \dots, z_d]$. If $d \geq 1$, the polynomial ring has dimension of at least one, and then R has dimension of at least one, a contradiction. Thus, $d = 0$, and R is module-finite over K . Since R is a field, this means precisely that R is a finite algebraic extension of K . *Q.E.D.*

Corollary

Let K be an algebraically closed field, let R be a finitely generated K -algebra, and let m be a maximal ideal of R . Then the composite map $K \rightarrow R \rightarrow R/m$ is an isomorphism.

Proof. R/m is a finitely generated K -algebra, since R is, and it is a field. Thus, $K \rightarrow R/m$ gives a finite algebraic extension of K . Since K is algebraically closed, it has no proper algebraic extension, and so $K \rightarrow R/m$ must be an isomorphism. *Q.E.D.*

Corollary

Let $R = K[x_1, \dots, x_n]$ be a polynomial ring over an algebraically closed field K . Then every maximal ideal m of R is the kernel of a K -homomorphism $K[x_1, \dots, x_n] \rightarrow K$, and so is determined by the elements $\lambda_1, \dots, \lambda_n \in K$ to which $x_1, \dots, x_n$ map. This maximal ideal is the kernel of the evaluation map $f(x_1, \dots, x_n) \mapsto f(\lambda_1, \dots, \lambda_n)$. It may also be described as the ideal $(x_1 - \lambda_1, \dots, x_n - \lambda_n) R$

Hilbert's Nullstellensatz, weak form

Corollary

Let $R = K[x_1, \dots, x_n]$ be a polynomial ring over an algebraically closed field K . Then every maximal ideal m of R is the kernel of a K -homomorphism $K[x_1, \dots, x_n] \rightarrow K$, and so is determined by the elements $\lambda_1, \dots, \lambda_n \in K$ to which $x_1, \dots, x_n$ map. This maximal ideal is the kernel of the evaluation map $f(x_1, \dots, x_n) \mapsto f(\lambda_1, \dots, \lambda_n)$. It may also be described as the ideal $(x_1 - \lambda_1, \dots, x_n - \lambda_n)R$.

Proof. Since $\gamma : K \cong R/m$, the K -algebra map $R \rightarrow R/m$, composed with γ^{-1} , gives a map $R \rightarrow K$ whose kernel is m . *Q.E.D.*

Hilbert's Nullstellensatz, weak form

Corollary

Let $R = K[x_1, \dots, x_n]$ be a polynomial ring over an algebraically closed field K . Then every maximal ideal m of R is the kernel of a K -homomorphism $K[x_1, \dots, x_n] \rightarrow K$, and so is determined by the elements $\lambda_1, \dots, \lambda_n \in K$ to which $x_1, \dots, x_n$ map. This maximal ideal is the kernel of the evaluation map $f(x_1, \dots, x_n) \mapsto f(\lambda_1, \dots, \lambda_n)$. It may also be described as the ideal $(x_1 - \lambda_1, \dots, x_n - \lambda_n)R$.

Proof. Since $\gamma : K \cong R/m$, the K -algebra map $R \rightarrow R/m$, composed with γ^{-1} , gives a map $R \rightarrow K$ whose kernel is m . *Q.E.D.*

Thus, when K is algebraically closed, we have a bijection between the points of K^n and the maximal ideals of $K[x_1, \dots, x_n]$.

Corollary

Let $f_1, \dots, f_n$ be polynomials in $K[x_1, \dots, x_n]$, where K is algebraically closed. Then the f_i generate the unit ideal (i.e., we have $1 = \sum_t g_t f_t$ for suitable polynomials g_t) if and only if the polynomials f_i do not vanish simultaneously, i.e., if and only if the algebraic set $V(f_1, \dots, f_n) = \emptyset$.

Corollary

Let $f_1, \dots, f_n$ be polynomials in $K[x_1, \dots, x_n]$, where K is algebraically closed. Then the f_i generate the unit ideal (i.e., we have $1 = \sum_t g_t f_t$ for suitable polynomials g_t) if and only if the polynomials f_i do not vanish simultaneously, i.e., if and only if the algebraic set $V(f_1, \dots, f_n) = \emptyset$.

Proof. If the f_i do not generate the unit ideal, the ideal they generate is contained in some maximal ideal of $K[x_1, \dots, x_n]$. But the functions in that maximal ideal all vanish at one point of K^n , a contradiction. On the other hand, if the f_i all vanish simultaneously at a point of K^n , they are in the maximal ideal of polynomials that vanish at that point: this direction does not need that K is algebraically closed. *Q.E.D.*

1 Finite Ring Extension

- Finite and Integral A -algebras
- Finite vs Integral
- Integral Closure

2 Noether Normalization

- Preliminary Results
- Noether Normalization Lemma
- Consequences of the Theorem
- Algebraic and Geometric Interpretation

Let's stress that **The Noether Normalization Lemma** is a fundamental result in: **Commutative Algebra** and **Algebraic Geometry** Let's look at the theorem from two key perspectives:

- 1 **Algebraic** Perspective
- 2 **Geometric** Perspective

Algebraic Point of View:

Algebraic Point of View:

- Even if a commutative algebra is complicated, there is always a simple **core**.

Algebraic Point of View:

- Even if a commutative algebra is complicated, there is always a simple **core**.
- This core is a **polynomial ring** in d variables, where d is the dimension of the algebra.

Algebraic Point of View:

- Even if a commutative algebra is complicated, there is always a simple **core**.
- This core is a **polynomial ring** in d variables, where d is the dimension of the algebra.
- The entire algebra can be described by adding a **finite number of additional generators**.

Algebraic and Geometric Interpretation

Algebraic Point of View:

- Even if a commutative algebra is complicated, there is always a simple **core**.
- This core is a **polynomial ring** in d variables, where d is the dimension of the algebra.
- The entire algebra can be described by adding a **finite number of additional generators**.

Geometric Point of View:

Algebraic Point of View:

- Even if a commutative algebra is complicated, there is always a simple **core**.
- This core is a **polynomial ring** in d variables, where d is the dimension of the algebra.
- The entire algebra can be described by adding a **finite number of additional generators**.

Geometric Point of View:

- Think of A as the **ring of functions** on an affine variety X .

Algebraic Point of View:

- Even if a commutative algebra is complicated, there is always a simple **core**.
- This core is a **polynomial ring** in d variables, where d is the dimension of the algebra.
- The entire algebra can be described by adding a **finite number of additional generators**.

Geometric Point of View:

- Think of A as the **ring of functions** on an affine variety X .
- Noether's theorem says: We can always find a **projection** onto an affine space k^d , that is a **finite map**.

Algebraic Point of View:

- Even if a commutative algebra is complicated, there is always a simple **core**.
- This core is a **polynomial ring** in d variables, where d is the dimension of the algebra.
- The entire algebra can be described by adding a **finite number of additional generators**.

Geometric Point of View:

- Think of A as the **ring of functions** on an affine variety X .
- Noether's theorem says: We can always find a **projection** onto an affine space k^d , that is a **finite map**.
- A **finite map** means there are finitely many points over each point of the base.

Algebraic and Geometric Interpretation

Algebraic Point of View:

- Even if a commutative algebra is complicated, there is always a simple **core**.
- This core is a **polynomial ring** in d variables, where d is the dimension of the algebra.
- The entire algebra can be described by adding a **finite number of additional generators**.

Geometric Point of View:

- Think of A as the **ring of functions** on an affine variety X .
- Noether's theorem says: We can always find a **projection** onto an affine space k^d , that is a **finite map**.
- A **finite map** means there are finitely many points over each point of the base.



KEY FOR OUR HOMOTOPY CONTINUATION METHOD

PART 2: THE APPLICATION

POSITIVE DIMENSIONAL SOLUTIONS TO POLYNOMIAL SYSTEMS VIA HOMOTOPY CONTINUATION

3 Homotopy Continuation Method

- Positive Dimensional Solutions
- Global Sampling via Linear Slicing
- Local Dimension Test
- Numerical Irreducible Decomposition via Monodromy

4 Numerical Computations

- Examples on Julia

3 Homotopy Continuation Method

- Positive Dimensional Solutions
 - Global Sampling via Linear Slicing
 - Local Dimension Test
 - Numerical Irreducible Decomposition via Monodromy

4 Numerical Computations

- Examples on Julia

Positive Dimensional Solutions

So far we have spoken about finding isolated solutions of a polynomial system $P(\mathbf{x}) = (p_1(\mathbf{x}), \dots, p_m(\mathbf{x}))$. We now want to delve deeper and investigate the positive dimensional solution set of $P(\mathbf{x})$.

Positive Dimensional Solutions

So far we have spoken about finding isolated solutions of a polynomial system $P(\mathbf{x}) = (p_1(\mathbf{x}), \dots, p_m(\mathbf{x}))$. We now want to delve deeper and investigate the positive dimensional solution set of $P(\mathbf{x})$.

- First, recall that the solution set of $P(\mathbf{x})$ can always be decomposed into a finite union of irreducible components.

So far we have spoken about finding isolated solutions of a polynomial system $P(\mathbf{x}) = (p_1(\mathbf{x}), \dots, p_m(\mathbf{x}))$. We now want to delve deeper and investigate the positive dimensional solution set of $P(\mathbf{x})$.

- First, recall that the solution set of $P(\mathbf{x})$ can always be decomposed into a finite union of irreducible components.
- Near almost all points in an irreducible component, the component locally looks like a neighbourhood of a smooth manifold of a fixed dimension. This dimension is the **dimension of the component**, and the collection of all such points is known as the smooth part of the component.

Positive Dimensional Solutions

So far we have spoken about finding isolated solutions of a polynomial system $P(\mathbf{x}) = (p_1(\mathbf{x}), \dots, p_m(\mathbf{x}))$. We now want to delve deeper and investigate the positive dimensional solution set of $P(\mathbf{x})$.

- First, recall that the solution set of $P(\mathbf{x})$ can always be decomposed into a finite union of irreducible components.
- Near almost all points in an irreducible component, the component locally looks like a neighbourhood of a smooth manifold of a fixed dimension. This dimension is the **dimension of the component**, and the collection of all such points is known as the smooth part of the component.
- We want to understand this decomposition via numerical methods, especially homotopy-based.

Consider the following square system of polynomials in the variables x, y, z :

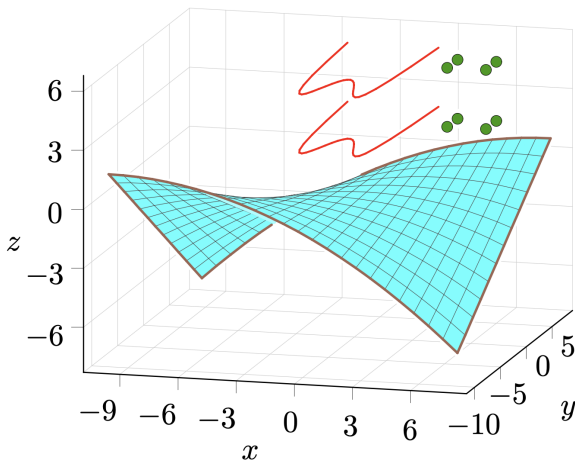
$$F(x, y, z) = \begin{bmatrix} f(x, y, z)g(x, y, z)(x - 4)(x - 6) \\ f(x, y, z)g(x, y, z)(y - 3)(y - 5) \\ f(x, y, z)(z - 2)(z - 5) \end{bmatrix}$$

where

$$f(x, y, z) = \frac{1}{40} (2xy - x^2) - z - 1 \quad \text{and} \quad g(x, y, z) = x^4 - 4x^2 - y - 1$$

Positive Dimensional Solutions

The following figure shows the real part of the variety V of $F(x, y, z) = 0$, consisting of a quadric (degree 2) surface, two quartic (degree 4) curves (at $z = 2$ and $z = 5$, respectively), and eight points. The surface is in blue, the two curves in red, and the eight points in green.



- ? Does a positive-dimensional solution set of $P(\mathbf{x}) = \mathbf{0}$ exist on a global scale?
If there is, can one obtain some sample points from each component?

- ? Does a positive-dimensional solution set of $P(\mathbf{x}) = \mathbf{0}$ exist on a global scale?
If there is, can one obtain some sample points from each component?
- ?? Given a solution $\mathbf{x} = \hat{\mathbf{x}} \in \mathbb{C}^n$ of the polynomial system $P(\mathbf{x}) = \mathbf{0}$, is $\hat{\mathbf{x}}$ an isolated solution? or does it belong to a positive dimensional solution component?

- ? Does a positive-dimensional solution set of $P(\mathbf{x}) = \mathbf{0}$ exist on a global scale?
If there is, can one obtain some sample points from each component?
- ?? Given a solution $\mathbf{x} = \hat{\mathbf{x}} \in \mathbb{C}^n$ of the polynomial system $P(\mathbf{x}) = \mathbf{0}$, is $\hat{\mathbf{x}}$ an isolated solution? or does it belong to a positive dimensional solution component?
- ??? Finally, can the global structure of the solution set of a polynomial system (ignoring the nonreduced structure) be studied via "numerical irreducible decomposition"?

3 Homotopy Continuation Method

- Positive Dimensional Solutions
- Global Sampling via Linear Slicing
- Local Dimension Test
- Numerical Irreducible Decomposition via Monodromy

4 Numerical Computations

- Examples on Julia

- **"Global sampling" of an algebraic set** $\rightarrow$ finding at least one sample point on *each* irreducible component.

Global Sampling via Linear Slicing

- **"Global sampling" of an algebraic set** $\rightarrow$ finding at least one sample point on *each* irreducible component.
- **Idea: Linear Slicing** $\rightarrow$ we intersect an irreducible component with an affine space of complementary dimension $\rightarrow$ we obtain isolated intersection points as sample points of the component.

Global Sampling via Linear Slicing

- **"Global sampling" of an algebraic set** $\rightarrow$ finding at least one sample point on *each* irreducible component.
- **Idea: Linear Slicing** $\rightarrow$ we intersect an irreducible component with an affine space of complementary dimension $\rightarrow$ we obtain isolated intersection points as sample points of the component.
- **Why does this work?** $\rightarrow$ **Noether Normalization Lemma:** any d -dimensional irreducible algebraic set in $\mathbb{C}^n$ can be realized as a "*finite branched cover*" over $\mathbb{C}^d$ via a linear projection.

Global Sampling via Linear Slicing

- **"Global sampling" of an algebraic set** $\rightarrow$ finding at least one sample point on *each* irreducible component.
- **Idea: Linear Slicing** $\rightarrow$ we intersect an irreducible component with an affine space of complementary dimension $\rightarrow$ we obtain isolated intersection points as sample points of the component.
- **Why does this work?** $\rightarrow$ **Noether Normalization Lemma:** any d -dimensional irreducible algebraic set in $\mathbb{C}^n$ can be realized as a "*finite branched cover*" over $\mathbb{C}^d$ via a linear projection.

NICE!

Baby Case: Curves

For a **1-dimensional algebraic set** (curve) in $\mathbb{C}^3$, this concept becomes more intuitive:

- Project the curve onto one of the coordinate axes, say the x -axis, using the projection $\pi(x, y, z) = x$.
- The projection map is a **finite-to-one** surjective map, meaning for almost all choices of x , the preimage $\pi^{-1}(x)$ contains finitely many points on the curve.
- More abstractly, via the projection π , the curve is realised as a finite branched cover over $\mathbb{C}$
- In this setup, by fixing an x and solving $\pi(\mathbf{p}) = x$, we obtain a **cross-section** of the curve.
- For almost all x , the number of complex points in $\pi^{-1}(x)$ is a fixed number, called the degree of the curve.

Problem:

- The process fails if the curve lies in a plane perpendicular to the x -axis, as the projection maps the entire curve to a single point, leaving other fibres empty.

Problem:

- The process fails if the curve lies in a plane perpendicular to the x -axis, as the projection maps the entire curve to a single point, leaving other fibres empty.

Solution:

- Noether's Lemma ensures that this failure does not occur when projecting onto a suitable subspace! This means that we can always find some one-dimensional subspace X of $\mathbb{C}^3$ such that π will be a surjective finite-to-one map over X .

Global Sampling via Linear Slicing

For a **d -dimensional** irreducible algebraic set $X \subset \mathbb{C}^n$:

- Consider linear projections to a d -dimensional subspace via a full-rank matrix A :

$$\pi(\mathbf{x}) = A\mathbf{x}$$

Global Sampling via Linear Slicing

For a **d-dimensional** irreducible algebraic set $X \subset \mathbb{C}^n$:

- Consider linear projections to a d -dimensional subspace via a full-rank matrix A :

$$\pi(\mathbf{x}) = A\mathbf{x}$$

- The map $\pi|_X : X \rightarrow \mathbb{C}^d$ restricts the projection to X , where for a point $\mathbf{b} \in \mathbb{C}^d$, the preimage $(\pi|_X)^{-1}(\mathbf{b})$ is a **linear slicing** of X .

Global Sampling via Linear Slicing

For a **d-dimensional** irreducible algebraic set $X \subset \mathbb{C}^n$:

- Consider linear projections to a d -dimensional subspace via a full-rank matrix A :

$$\pi(\mathbf{x}) = A\mathbf{x}$$

- The map $\pi|_X : X \rightarrow \mathbb{C}^d$ restricts the projection to X , where for a point $\mathbf{b} \in \mathbb{C}^d$, the preimage $(\pi|_X)^{-1}(\mathbf{b})$ is a **linear slicing** of X .
- By **Noether's Normalization Lemma**, for generic choices of the matrix A and point $\mathbf{b} \in \mathbb{C}^d$:

Global Sampling via Linear Slicing

For a **d-dimensional** irreducible algebraic set $X \subset \mathbb{C}^n$:

- Consider linear projections to a d -dimensional subspace via a full-rank matrix A :

$$\pi(\mathbf{x}) = A\mathbf{x}$$

- The map $\pi|_X : X \rightarrow \mathbb{C}^d$ restricts the projection to X , where for a point $\mathbf{b} \in \mathbb{C}^d$, the preimage $(\pi|_X)^{-1}(\mathbf{b})$ is a **linear slicing** of X .
- By **Noether's Normalization Lemma**, for generic choices of the matrix A and point $\mathbf{b} \in \mathbb{C}^d$:
 - $(\pi|_X)^{-1}(\mathbf{b})$ consists of a **finite number of points** in the smooth part X_{smooth} of X .
 - ? For almost all $\mathbf{b}$, the number of points is fixed, called the **degree** of X , denoted as $\deg X$.

Global Sampling via Linear Slicing

Notice that $(\pi|_X)^{-1}(\mathbf{b})$ is the set of intersection points between X and the $(n - d)$ -dimensional affine space defined by $A\mathbf{x} = \mathbf{b}$, and is called a generic linear slicing of X .

Global Sampling via Linear Slicing

Notice that $(\pi|_X)^{-1}(\mathbf{b})$ is the set of intersection points between X and the $(n - d)$ -dimensional affine space defined by $A\mathbf{x} = \mathbf{b}$, and is called a generic linear slicing of X .

Algebraically, if X is an irreducible component of the solution set of the polynomial system $P(\mathbf{x}) = \mathbf{0}$, then $(\pi|_X)^{-1}(\mathbf{b})$ is a subset of the solution set of the augmented system $(P(\mathbf{x}), A\mathbf{x} - \mathbf{b})$, that is,

$$\left\{ \begin{array}{l} p_1(x_1, \dots, x_n) = 0 \\ \vdots \\ p_m(x_1, \dots, x_n) = 0 \\ a_{11}x_1 + \dots + a_{1n}x_n = b_1 \\ \vdots \\ a_{d1}x_1 + \dots + a_{dn}x_n = b_d \end{array} \right. \quad (*)$$

Global Sampling via Linear Slicing

Notice that $(\pi|_X)^{-1}(\mathbf{b})$ is the set of intersection points between X and the $(n - d)$ -dimensional affine space defined by $A\mathbf{x} = \mathbf{b}$, and is called a generic linear slicing of X .

Algebraically, if X is an irreducible component of the solution set of the polynomial system $P(\mathbf{x}) = \mathbf{0}$, then $(\pi|_X)^{-1}(\mathbf{b})$ is a subset of the solution set of the augmented system $(P(\mathbf{x}), A\mathbf{x} - \mathbf{b})$, that is,

$$\left\{ \begin{array}{l} p_1(x_1, \dots, x_n) = 0 \\ \vdots \\ p_m(x_1, \dots, x_n) = 0 \\ a_{11}x_1 + \dots + a_{1n}x_n = b_1 \\ \vdots \\ a_{d1}x_1 + \dots + a_{dn}x_n = b_d \end{array} \right. \quad (*)$$



Finite many solutions → **Classical Homotopy Continuation**

Global Sampling via Linear Slicing: Algorithm

Step 1: For the given target dimension d , pick a random $d \times n$ matrix $A \in M_{d \times n}(\mathbb{C})$ and a random vector $\mathbf{b} \in \mathbb{C}^d$. With these, one constructs the augmented "linear slicing" system.

Global Sampling via Linear Slicing: Algorithm

Step 1: For the given target dimension d , pick a random $d \times n$ matrix $A \in M_{d \times n}(\mathbb{C})$ and a random vector $\mathbf{b} \in \mathbb{C}^d$. With these, one constructs the augmented "linear slicing" system.

Step 2: Solve the augmented system via numerical homotopy methods, possibly in conjunction with randomization techniques if it is not square.

Global Sampling via Linear Slicing: Algorithm

Step 1: For the given target dimension d , pick a random $d \times n$ matrix $A \in M_{d \times n}(\mathbb{C})$ and a random vector $\mathbf{b} \in \mathbb{C}^d$. With these, one constructs the augmented "linear slicing" system.

Step 2: Solve the augmented system via numerical homotopy methods, possibly in conjunction with randomization techniques if it is not square.

Step 3: Apply local dimension test techniques to the solutions of the system to obtain and filter out nonisolated solutions. The remaining solutions contain at least one point on each irreducible d -dimensional component of the solution set of the original system $P(\mathbf{x}) = \mathbf{0}$.

Global Sampling via Linear Slicing

Additional sample points can be generated by moving the linear slicing. That is, one could consider the one-parameter family of linear slicings given by $A_t \mathbf{x} = \mathbf{b}_t$ where A_0 and $\mathbf{b}_0$ represents the linear slicing (*) and the homotopy

$$H(\mathbf{x}, t) := \begin{cases} P(\mathbf{x}) \\ A_t \mathbf{x} - \mathbf{b}_t \end{cases}$$

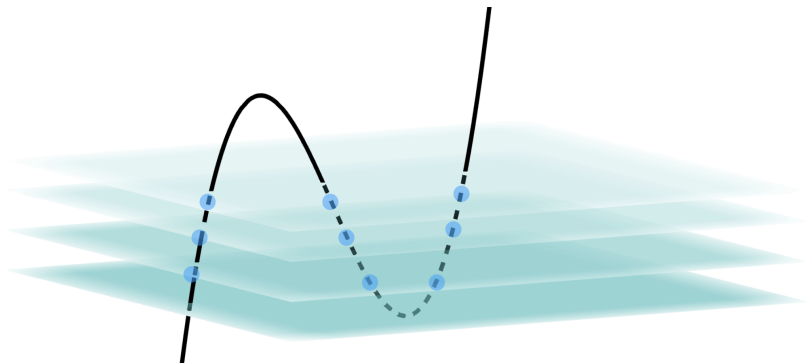
The sample points obtained by solving (*) are isolated nonsingular solutions of $H(\mathbf{x}, 0) = \mathbf{0}$ and can thus be used as the starting points of the homotopy paths of this homotopy.

Global Sampling via Linear Slicing

Additional sample points can be generated by moving the linear slicing. That is, one could consider the one-parameter family of linear slicings given by $A_t \mathbf{x} = \mathbf{b}_t$ where A_0 and $\mathbf{b}_0$ represents the linear slicing (*) and the homotopy

$$H(\mathbf{x}, t) := \begin{cases} P(\mathbf{x}) \\ A_t \mathbf{x} - \mathbf{b}_t \end{cases}$$

The sample points obtained by solving (*) are isolated nonsingular solutions of $H(\mathbf{x}, 0) = \mathbf{0}$ and can thus be used as the starting points of the homotopy paths of this homotopy.



Global Sampling via Linear Slicing

The (numerical approximations of the) point we obtain are called a **witness set**.

Definition

Suppose that $F(\mathbf{x})$ is a system of polynomials, and let V be an irreducible component of the variety defined by $F(\mathbf{x}) = 0$.

Global Sampling via Linear Slicing

The (numerical approximations of the) point we obtain are called a **witness set**.

Definition

Suppose that $F(\mathbf{x})$ is a system of polynomials, and let V be an irreducible component of the variety defined by $F(\mathbf{x}) = 0$.

A witness set W for the component V is a triple $W = (F, L, L \cap V)$, where L is a general (affine) linear subspace complementary to V in that $\text{codim}(L) = \dim(V)$ and $L \cap V$ consists of numerical approximations of the points in the intersection of L and V .

Global Sampling via Linear Slicing

The (numerical approximations of the) point we obtain are called a **witness set**.

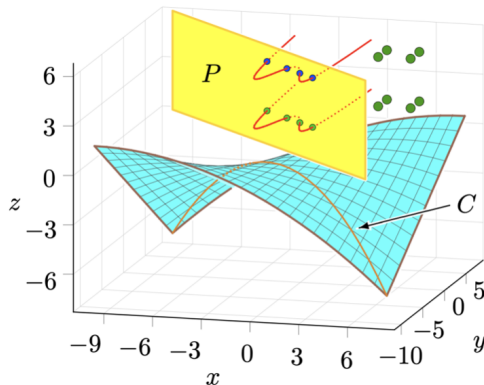
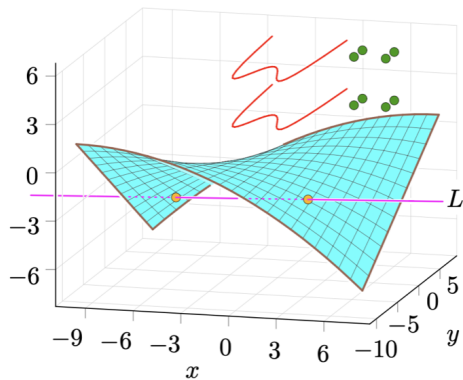
Definition

Suppose that $F(\mathbf{x})$ is a system of polynomials, and let V be an irreducible component of the variety defined by $F(\mathbf{x}) = 0$.

A witness set W for the component V is a triple $W = (F, L, L \cap V)$, where L is a general (affine) linear subspace complementary to V in that $\text{codim}(L) = \dim(V)$ and $L \cap V$ consists of numerical approximations of the points in the intersection of L and V .

In practice, L is represented by $\dim(V)$ randomly chosen polynomials of degree one.

Global Sampling via Linear Slicing



3 Homotopy Continuation Method

- Positive Dimensional Solutions
- Global Sampling via Linear Slicing
- Local Dimension Test
- Numerical Irreducible Decomposition via Monodromy

4 Numerical Computations

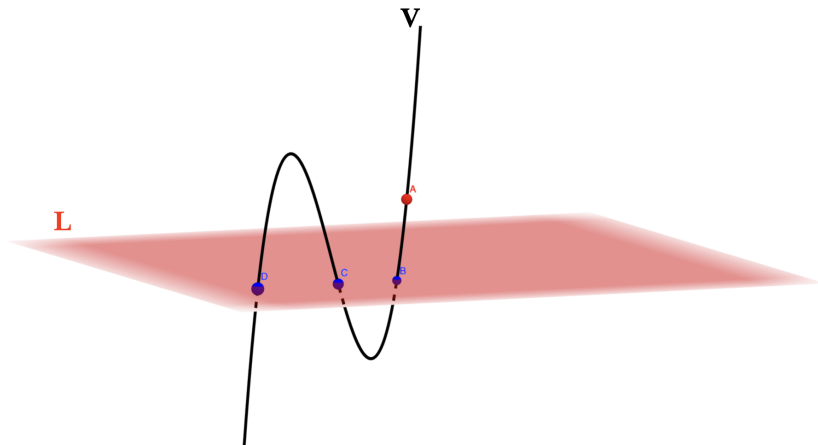
- Examples on Julia

How do we know if a solution point is isolated or belongs to a positive dimensional solution component?

Local Dimension Test

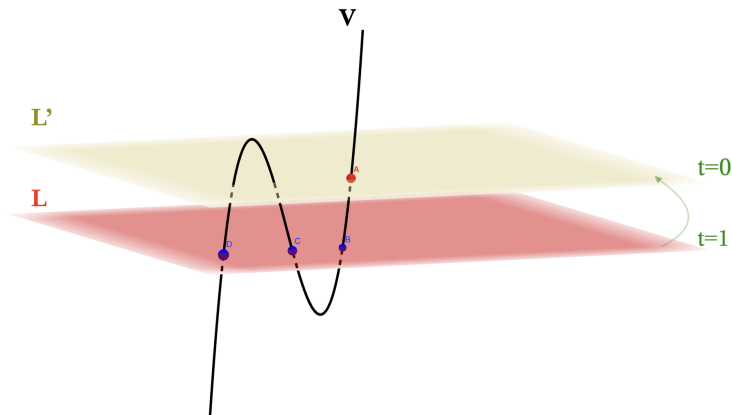
How do we know if a solution point is isolated or belongs to a positive dimensional solution component?

Let $(F, L, L \cap V)$ be a witness set for $V \in \mathbb{C}^n$. Let $A \in \mathbb{C}^n$.
We want to determine if $A \in V$.



Local Dimension Test: Algorithm

Step 1:



$$H(\mathbf{x}; t) = (f(\mathbf{x}), tL(\mathbf{x}) + (1 - t)L'(\mathbf{x}))$$

$L' =$ linear affine space passing through A .

Step 2: To compute the dimension of the component, I can use the rank of the Jacobian evaluated on a point of a component (SVD or reduced SVD, if needed)

Local Dimension Test

- Recall that, every time we are talking about homotopy continuation, we're talking about a polynomial system with finitely many solutions, moving over a parameter space of polynomial systems.

Local Dimension Test

- Recall that, every time we are talking about homotopy continuation, we're talking about a polynomial system with finitely many solutions, moving over a parameter space of polynomial systems.
- Here, the idea is that I have my variety V , I have added to the system the linear forms defining the plane, hence I have a polynomial system with finitely many points.

Local Dimension Test

- Recall that, every time we are talking about homotopy continuation, we're talking about a polynomial system with finitely many solutions, moving over a parameter space of polynomial systems.
- Here, the idea is that I have my variety V , I have added to the system the linear forms defining the plane, hence I have a polynomial system with finitely many points.
- The parameter space is the space of affine linear spaces of dimension equal to the codimension of V : $G(n - d, n)$.

Local Dimension Test

- Recall that, every time we are talking about homotopy continuation, we're talking about a polynomial system with finitely many solutions, moving over a parameter space of polynomial systems.
- Here, the idea is that I have my variety V , I have added to the system the linear forms defining the plane, hence I have a polynomial system with finitely many points.
- The parameter space is the space of affine linear spaces of dimension equal to the codimension of V : $G(n - d, n)$.
- Recall that we can do homotopy if in the parameter space there is a Zariski-open (dense) subset in which all the points correspond to linear systems with (the maximal number of) finitely many solutions.

Local Dimension Test

- Recall that, every time we are talking about homotopy continuation, we're talking about a polynomial system with finitely many solutions, moving over a parameter space of polynomial systems.
- Here, the idea is that I have my variety V , I have added to the system the linear forms defining the plane, hence I have a polynomial system with finitely many points.
- The parameter space is the space of affine linear spaces of dimension equal to the codimension of V : $G(n - d, n)$.
- Recall that we can do homotopy if in the parameter space there is a Zariski-open (dense) subset in which all the points correspond to linear systems with (the maximal number of) finitely many solutions.
- We can always do homotopy by "staying in that region", i.e. by avoiding the subvariety that contains polynomial systems with fewer solutions.

Local Dimension Test

- Recall that, every time we are talking about homotopy continuation, we're talking about a polynomial system with finitely many solutions, moving over a parameter space of polynomial systems.
- Here, the idea is that I have my variety V , I have added to the system the linear forms defining the plane, hence I have a polynomial system with finitely many points.
- The parameter space is the space of affine linear spaces of dimension equal to the codimension of V : $G(n - d, n)$.
- Recall that we can do homotopy if in the parameter space there is a Zariski-open (dense) subset in which all the points correspond to linear systems with (the maximal number of) finitely many solutions.
- We can always do homotopy by "staying in that region", i.e. by avoiding the subvariety that contains polynomial systems with fewer solutions.
- **Bertini's Theorem:** There exists a dense Zariski-open U subset such that for any $L \in U$, $L \cap X$ consists only of simple points and the cardinality is the largest possible ($= \deg(X)$).

3 Homotopy Continuation Method

- Positive Dimensional Solutions
- Global Sampling via Linear Slicing
- Local Dimension Test
- Numerical Irreducible Decomposition via Monodromy

4 Numerical Computations

- Examples on Julia

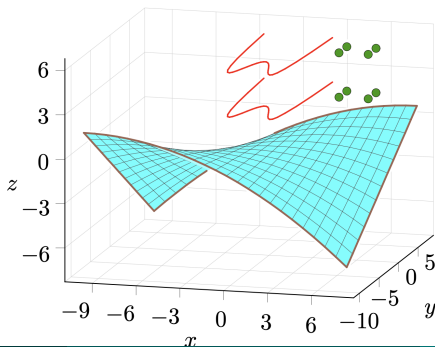
Numerical Irreducible Decomposition via Monodromy

Given an algebraic set $\mathcal{V}(P) = \{\mathbf{x} \in \mathbb{C}^n \mid P(\mathbf{x}) = \mathbf{0}\}$, we aim to find a *numerical irreducible decomposition* that mirrors the theoretical one.

Specifically, we express $\mathcal{V}(P)$ as:

$$\mathcal{V}(P) = \bigcup_{d=0}^n V_d = \bigcup_{d=0}^n \bigcup_{i \in I_d} V_{d,i}$$

where $V_{d,i}$ is an irreducible component of dimension d , and I_d is a finite index set. Our goal is to numerically decompose these components, starting with sets of pure dimension.



Numerical Irreducible Decomposition via Monodromy

- Since components with positive dimensions contain infinitely many points, we need a finite, numerically computable encoding.

We again use **witness sets**. $(F, L_z, L_z \cap V_d)$

Numerical Irreducible Decomposition via Monodromy

- Since components with positive dimensions contain infinitely many points, we need a finite, numerically computable encoding.
We again use **witness sets**. $(F, L_z, L_z \cap V_d)$
- To partition the witness set into subsets corresponding to different irreducible components, we use the **monodromy technique**.

Numerical Irreducible Decomposition via Monodromy

- Since components with positive dimensions contain infinitely many points, we need a finite, numerically computable encoding.
We again use **witness sets**. $(F, L_Z, L_Z \cap V_d)$
- To partition the witness set into subsets corresponding to different irreducible components, we use the **monodromy technique**.
- The smooth part of each irreducible component is path-connected, while the smooth parts of different components are disjoint.

Numerical Irreducible Decomposition via Monodromy

- Since components with positive dimensions contain infinitely many points, we need a finite, numerically computable encoding.
We again use **witness sets**. $(F, L_Z, L_Z \cap V_d)$
- To partition the witness set into subsets corresponding to different irreducible components, we use the **monodromy technique**.
- The smooth part of each irreducible component is path-connected, while the smooth parts of different components are disjoint.
- Monodromy constructs paths in the smooth parts of components, connecting different witness points within the same component.

Numerical Irreducible Decomposition via Monodromy

- Since components with positive dimensions contain infinitely many points, we need a finite, numerically computable encoding.
We again use **witness sets**. $(F, L_z, L_z \cap V_d)$
- To partition the witness set into subsets corresponding to different irreducible components, we use the **monodromy technique**.
- The smooth part of each irreducible component is path-connected, while the smooth parts of different components are disjoint.
- Monodromy constructs paths in the smooth parts of components, connecting different witness points within the same component.
- A family of linear slicings $L_z(\mathbf{x})$ parametrized by $z \in \mathbb{C}$ moves witness points smoothly, except at special *branch points*.
Key: $U = \{z \in \mathbb{C} \mid L_z(\mathbf{x}) \text{ is a generic linear slicing wrt } V_d\}$ is dense in $\mathbb{C}$.
 U^c may contain isolated "branch points": as z moves across them, some witness points in W_z may collide.

Numerical Irreducible Decomposition via Monodromy

- Since components with positive dimensions contain infinitely many points, we need a finite, numerically computable encoding.
We again use **witness sets**. $(F, L_z, L_z \cap V_d)$
- To partition the witness set into subsets corresponding to different irreducible components, we use the **monodromy technique**.
- The smooth part of each irreducible component is path-connected, while the smooth parts of different components are disjoint.
- Monodromy constructs paths in the smooth parts of components, connecting different witness points within the same component.
- A family of linear slicings $L_z(\mathbf{x})$ parametrized by $z \in \mathbb{C}$ moves witness points smoothly, except at special *branch points*.

Key: $U = \{z \in \mathbb{C} \mid L_z(\mathbf{x}) \text{ is a generic linear slicing wrt } V_d\}$ is dense in $\mathbb{C}$.

U^c may contain isolated "branch points": as z moves across them, some witness points in W_z may collide.

- By tracking the movement of witness points around a branch point, we can reveal nontrivial **monodromy**, connecting witness points in the same irreducible component.

Key: I can always **smoothly** transition from one point to another of the same irreducible component ($\mathbb{C}!$)

Example: Quadratic Curve

Consider the equation $P(x, y) = x^2 - y = 0$, which defines a parabola in $\mathbb{C}^2$. Let $z \in \mathbb{C}$. Using the linear slicing $L_z(x, y) = y - z = 0$, the system becomes:

$$x^2 - y = 0$$

$$y - z = 0$$

Example: Quadratic Curve

Consider the equation $P(x, y) = x^2 - y = 0$, which defines a parabola in $\mathbb{C}^2$. Let $z \in \mathbb{C}$. Using the linear slicing $L_z(x, y) = y - z = 0$, the system becomes:

$$x^2 - y = 0$$

$$y - z = 0$$

For $z \neq 0$, there are two witness points: $(x, y) = (\pm\sqrt{z}, z)$.

As $z \rightarrow 0$, these points collide at the branch point $z = 0$.

Example: Quadratic Curve

Consider the equation $P(x, y) = x^2 - y = 0$, which defines a parabola in $\mathbb{C}^2$. Let $z \in \mathbb{C}$. Using the linear slicing $L_z(x, y) = y - z = 0$, the system becomes:

$$x^2 - y = 0$$

$$y - z = 0$$

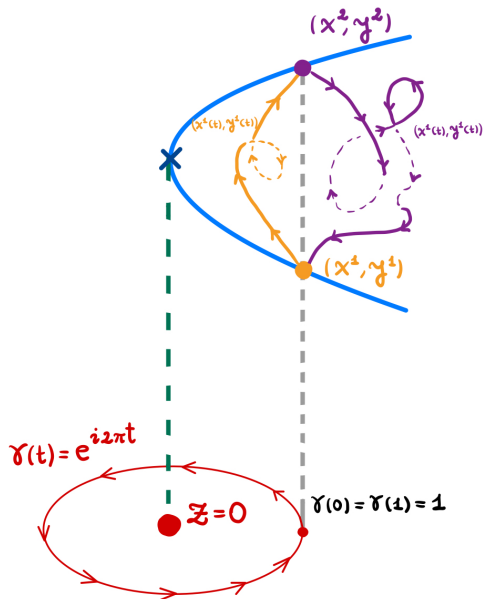
For $z \neq 0$, there are two witness points: $(x, y) = (\pm\sqrt{z}, z)$.

As $z \rightarrow 0$, these points collide at the branch point $z = 0$. By looping $z = \gamma(t) = e^{i2\pi t}$ around the origin, we track how the witness points exchange positions, revealing nontrivial monodromy and showing that the points belong to the same irreducible component.

Example: Quadratic Curve

For $t \in [0, 1]$:

- $(x^{(1)}(t), y^{(1)}(t)) = (e^{i\pi t}, e^{i2\pi t})$
- $(x^{(2)}(t), y^{(2)}(t)) = (-e^{i\pi t}, e^{i2\pi t})$



Numerical Irreducible Decomposition via Monodromy: Algorithm

Step 1: For a fixed dimension d , let W_0 be the witness set of V_d in obtained by solving the augmented system using homotopy methods.

Numerical Irreducible Decomposition via Monodromy: Algorithm

Step 1: For a fixed dimension d , let W_0 be the witness set of V_d in obtained by solving the augmented system using homotopy methods.

Step 2: A one-parameter family of linear slicings $L_z(\mathbf{x}) : \mathbb{C}^n \rightarrow \mathbb{C}^d, z \in \mathbb{C}$ is constructed with the property that for a generic choice of z , $L(\mathbf{x}) := L_z(\mathbf{x})$ defines a linear slicing that is generic wrt the components of V_d . Within the set of "generic choices", a simple loop $\gamma : [0, 1] \rightarrow \mathbb{C}$ is chosen.

Numerical Irreducible Decomposition via Monodromy: Algorithm

Step 1: For a fixed dimension d , let W_0 be the witness set of V_d in obtained by solving the augmented system using homotopy methods.

Step 2: A one-parameter family of linear slicings $L_z(\mathbf{x}) : \mathbb{C}^n \rightarrow \mathbb{C}^d, z \in \mathbb{C}$ is constructed with the property that for a generic choice of z , $L(\mathbf{x}) := L_z(\mathbf{x})$ defines a linear slicing that is generic wrt the components of V_d . Within the set of "generic choices", a simple loop $\gamma : [0, 1] \rightarrow \mathbb{C}$ is chosen.

Step 3: Using witness points in W_0 as starting points, one tracks the solution paths defined by

$$H(\mathbf{x}, t) := \begin{cases} P(\mathbf{x}) = \mathbf{0} \\ L_{\gamma(t)}(\mathbf{x}) = \mathbf{0} \end{cases}$$

for t from 0 to 1. The endpoints of these solution paths are necessarily inside the same witness set W_0 . Most importantly, when a solution path joins two different witness points, then these two points must be in the same irreducible components.

Numerical Irreducible Decomposition via Monodromy: Algorithm

Step 1: For a fixed dimension d , let W_0 be the witness set of V_d in obtained by solving the augmented system using homotopy methods.

Step 2: A one-parameter family of linear slicings $L_z(\mathbf{x}) : \mathbb{C}^n \rightarrow \mathbb{C}^d, z \in \mathbb{C}$ is constructed with the property that for a generic choice of z , $L(\mathbf{x}) := L_z(\mathbf{x})$ defines a linear slicing that is generic wrt the components of V_d . Within the set of "generic choices", a simple loop $\gamma : [0, 1] \rightarrow \mathbb{C}$ is chosen.

Step 3: Using witness points in W_0 as starting points, one tracks the solution paths defined by

$$H(\mathbf{x}, t) := \begin{cases} P(\mathbf{x}) = \mathbf{0} \\ L_{\gamma(t)}(\mathbf{x}) = \mathbf{0} \end{cases}$$

for t from 0 to 1. The endpoints of these solution paths are necessarily inside the same witness set W_0 . Most importantly, when a solution path joins two different witness points, then these two points must be in the same irreducible components.

Steps 2 and 3 can be repeated with a different family of slicings and potentially reveal the connectedness (via solution paths) between other witness points. The *trace test* can be used as a stopping criterion.

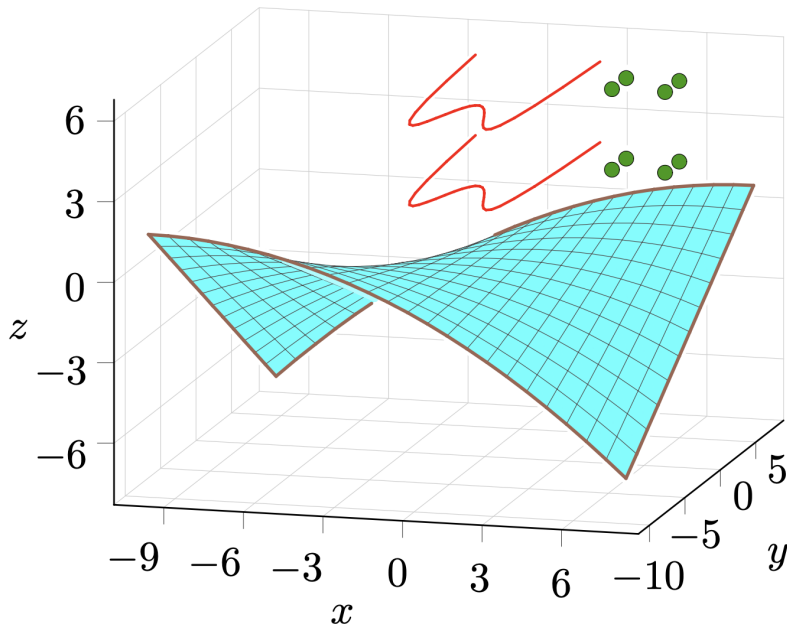
3 Homotopy Continuation Method

- Positive Dimensional Solutions
- Global Sampling via Linear Slicing
- Local Dimension Test
- Numerical Irreducible Decomposition via Monodromy

4 Numerical Computations

- Examples on Julia

Examples on Julia



References

-  T.-Y. Li, T.-R. Chen, “Numerical Algebraic Geometry and its Applications”, *SIAM Journal on Applied Algebra and Geometry*, 2018.
-  A.J. Sommese, C.W. Wampler, “The Numerical Solution of Systems of Polynomials”, *World Scientific Publishing*, 2005.
-  A. Oneto “Geometry and Topology for Data Analysis”, *University of Trento*, 2024.
-  A. Sommese, J. Verschelde, C. Wampler, “Numerical Irreducible Decomposition Using Monodromy”, *Journal of Symbolic Computation*, 2001.
-  B. Hauenstein, “Witness Sets and Homotopy Methods for Numerical Algebraic Geometry”, *Mathematics of Computation*, 2012.
-  Miles Reid, *Undergraduate Commutative Algebra*, Cambridge University Press, 1995.
-  T.-R. Chen, T.-Y. Li, “The Gamma-Trick and its Application in Numerical Decompositions”, *Mathematics of Computation*, 2005.

Thank you for your attention!